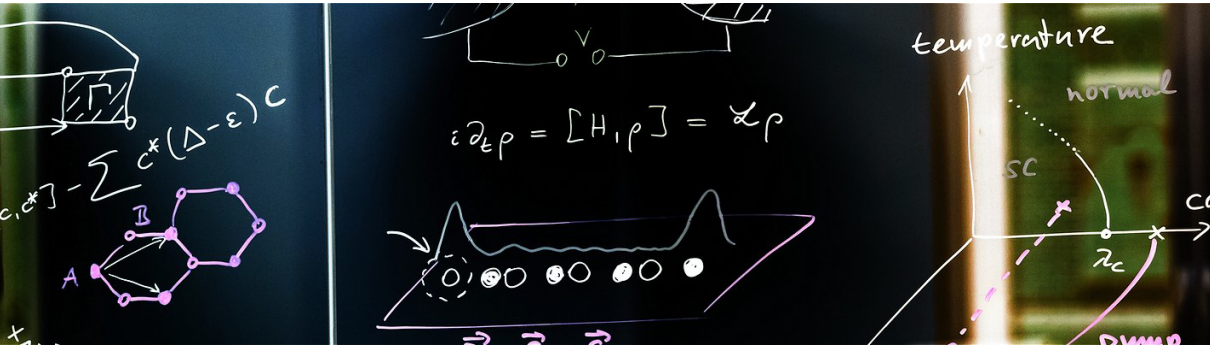


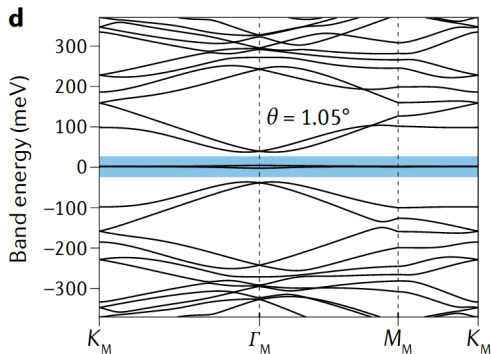


Tjark Sievers – Master's Colloquium

May 2025

# Superconducting Phase Stiffness and Coherence Length From a Finite Momentum Pairing Constraint

# Flat band Graphene systems

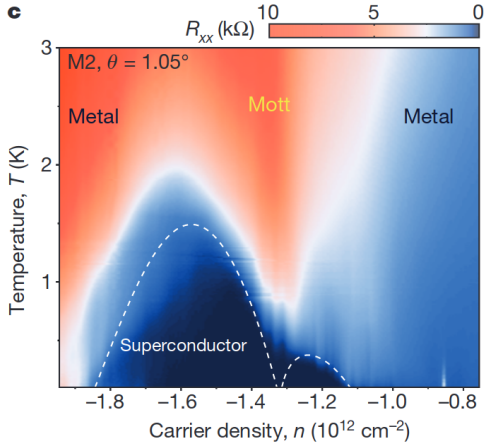


- ▶ Flat bands: quenched kinetic energy, enhanced interaction effects
- ▶ On the left: twisted Bilayer Graphene at the magic angle<sup>1</sup> (MATBG)
- ▶ Interesting topology due to hybridization of localized and extended topological orbitals (topological heavy fermion model<sup>2</sup>)

<sup>1</sup>Band structure taken from Törmä, Peotta, and Bernevig, [10.1038/s42254-022-00466-y](https://doi.org/10.1038/s42254-022-00466-y) (2022)

<sup>2</sup>Song and Bernevig [10.1103/PhysRevLett.129.047601](https://doi.org/10.1103/PhysRevLett.129.047601) (2022)

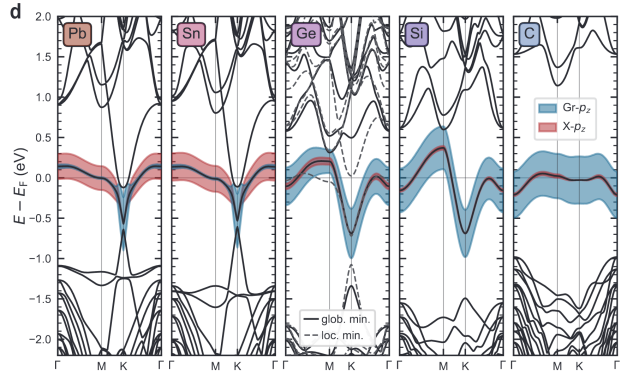
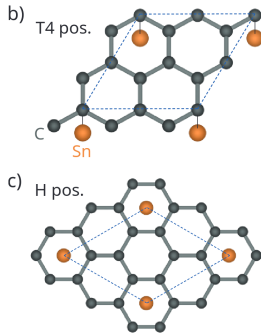
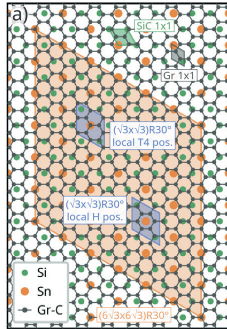
# Superconductivity<sup>3</sup>



► Superconductivity

<sup>3</sup>Cao et al., [10.1038/nature26160](https://doi.org/10.1038/nature26160) (2018)

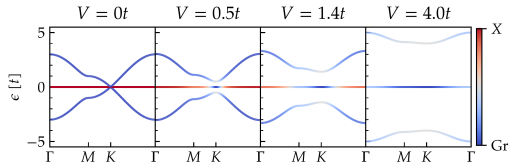
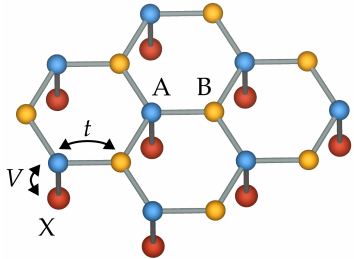
# Proximity Coupling<sup>45</sup>



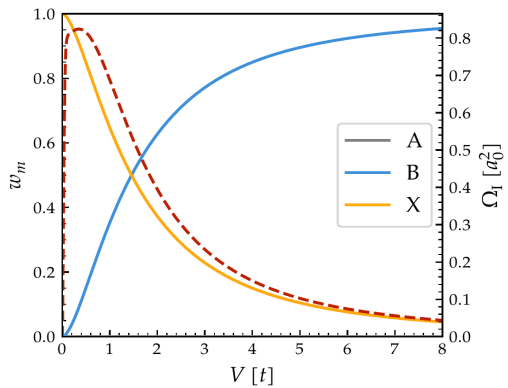
<sup>4</sup>Ghosal et al.[10.48550/arXiv.2412.01329](https://arxiv.org/abs/2412.01329) (2024)

<sup>5</sup>Witt et al.[10.48550/arXiv.2503.03700](https://arxiv.org/abs/2503.03700) (2025)

# Decorated Graphene model<sup>6</sup>



<sup>6</sup>Witt et al. [10.48550/arXiv.2503.03700](https://arxiv.org/abs/2503.03700) (2025)



# Ginzburg-Landau Theory

Two ingredients:

- Order parameter:

$$|\Psi| = \begin{cases} 0 & T \geq T_C \\ |\Psi_0| > 0 & T < T_C \end{cases} . \quad (1)$$

- Free energy: thermodynamic potential, minimized in thermodynamic equilibrium

- Ginzburg-Landau free energy (expansion of free energy in order parameter):

$$f_{\text{GL}}[\Psi, \nabla\Psi] = \frac{\hbar^2}{2m^*} |\nabla\Psi|^2 + r|\Psi|^2 + \frac{u}{2} |\Psi|^4 \quad (2)$$

- Writing  $\Psi = |\Psi|e^{i\phi}$ :

$$f_{\text{GL}} = \frac{\hbar^2}{2m^*} |\Psi|^2 (\nabla\phi)^2 + \left[ \frac{\hbar^2}{2m^*} (\nabla|\Psi|)^2 + r|\Psi|^2 + \frac{u}{2} |\Psi|^4 \right] \quad (3)$$

- Correlation length:

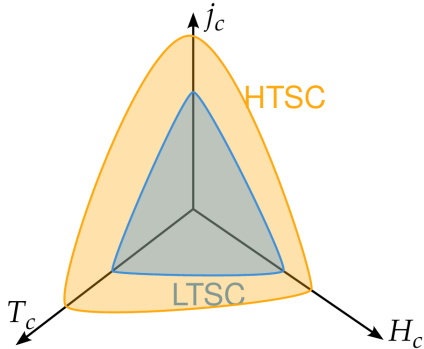
$$\xi = \sqrt{\frac{\hbar^2}{2m^*|r|}} = \xi_0 \left(1 - \frac{T}{T_C}\right)^{-\frac{1}{2}} \quad (4)$$

with coherence length  $\xi_0 = \xi(T = 0) = \sqrt{\frac{\hbar^2}{2maT_C}}$



- ▶ Coupling to electromagnetic field via vector potential  $\mathbf{A}$ , i.e.  $f_{\text{GL}}[\Psi, \mathbf{A}]$
- ▶ Another characteristic length scales for variations of  $\mathbf{A}$ : London penetration depth  $\lambda_{\text{L},0}$
- ▶

# Why are these length scales important?



- Critical fields and currents
- Characterize what is driving the pairing: superfluid weight

- ▶ Stationary point of Ginzburg-Landau energy:

$$|\Psi| = |\Psi_0| \sqrt{1 - \xi^2 |\nabla \phi(\mathbf{r})|^2} \quad (5)$$

- ▶ **Breakdown of order parameter with phase fluctuations**  $\rightarrow$  **correlation length  $\xi$**
- ▶ Easiest choice:
- ▶ Finite momentum  $\rightarrow$  current  $\mathbf{j}_q$ , can get

$$\lambda_{L,0} \quad (6)$$

Summary so far:

- ▶ Introduced length scales  $\lambda_{L,0}$  and coherence length  $\xi_0$
- ▶ Connect to critical fields and currents and can help characterize pairing
- ▶ Can get information on length scales from the breakdown of the order parameter with phase fluctuations

# FMP method in microscopic theories

## BCS theory

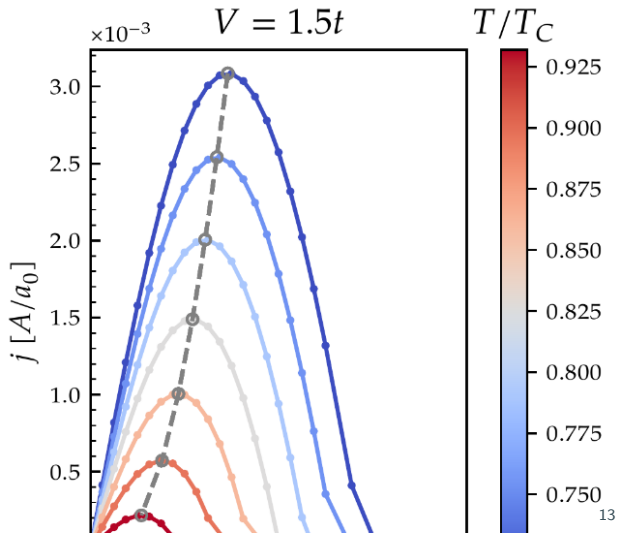
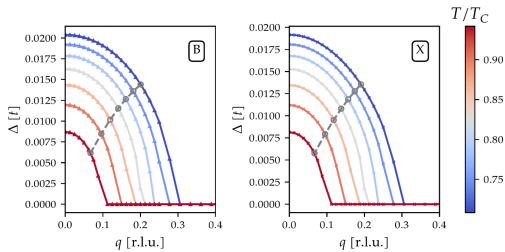
►  $H_{\text{int}} \approx \sum_{i,\alpha} \left( \Delta_{i\alpha} c_{i\alpha\uparrow}^\dagger c_{i\alpha\downarrow}^\dagger + \Delta_{i\alpha}^* c_{i\alpha\downarrow} c_{i\alpha\uparrow} \right)$

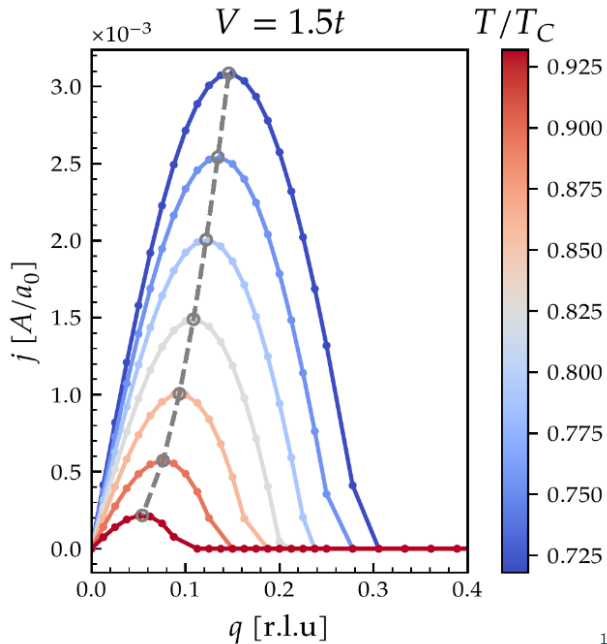
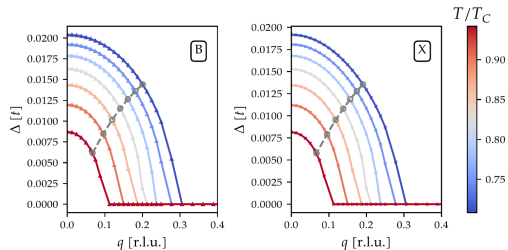


## DMFT



# Results: FMP Method in BCS Theory

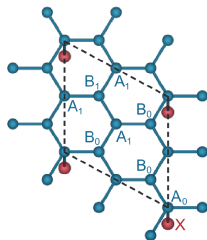




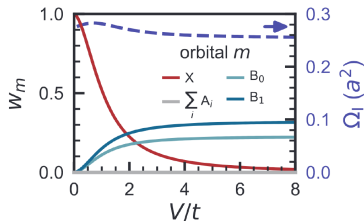
# Outlook

- ▶ Current project: benchmark FMP method in DMFT for one band Hubbard model
- ▶ Compare FMP method in BCS with other results in flat band systems (e.g. Lieb lattice<sup>7</sup> as a standard example), clear up the discrepancies
- ▶ Apply FMP method to (more realistic) extended decorated Graphene model

(a)



(b)



<sup>7</sup>Penttilä, Huhtinen, and Törmä 10.1038/s42005-025-01964-y (2025)



# Summary

- ▶ Showed
- ▶ Introduced finite-momentum pairing method to calculate superconducting length scales
- ▶